PCBC H Name _____ _____ /32

Limits Quiz 1.1–1.2 *Non-Calculator* You have 30 minutes.

PRACTICE B · built to match the real Quiz 1.1–1.2 (same layout, same question types)

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

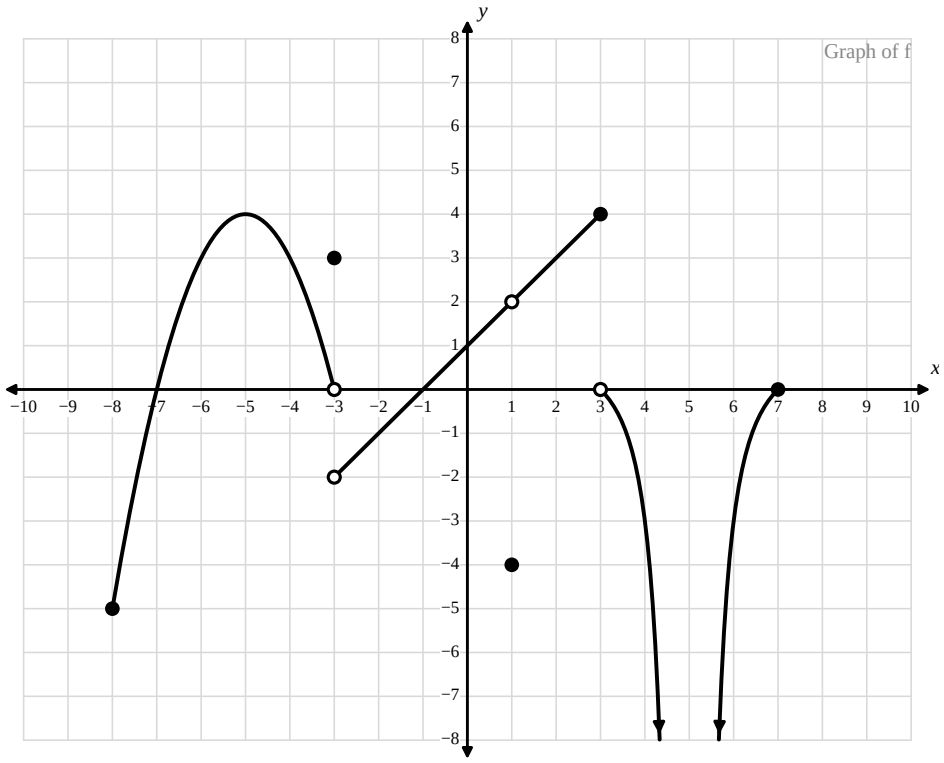
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (1 point each)

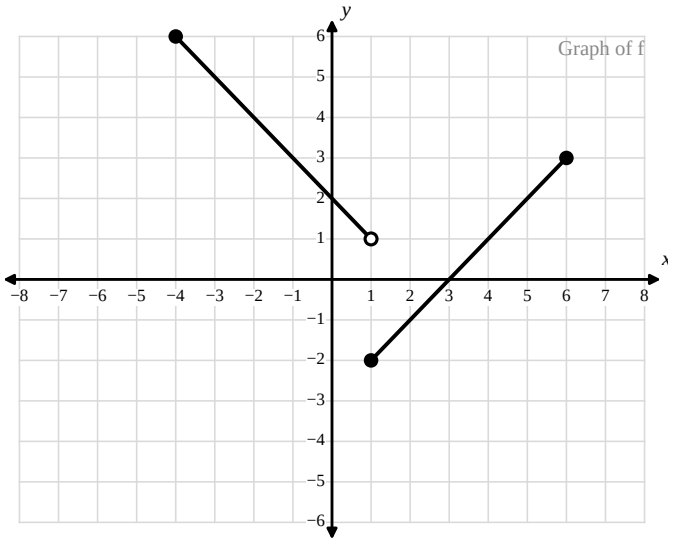


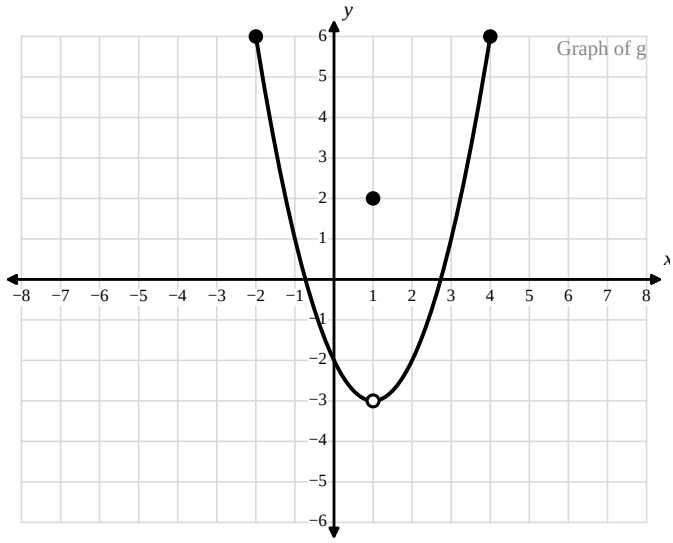
1. $\lim_{x\to 1} f(x) =$	**2.** $f(1) =$
3. $\lim_{x\to -3} f(x) =$	**4.** $\lim_{x\to 5} f(x) =$

(2 points)

5. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find each limit. Show the limit properties you use. (2 points each)





6. $\lim_{x \to 1^-} [f(x)\, g(x)]$

7. $\lim_{x \to 1} [f(x) + g(x)]$

(2 points)

8. Use the table to estimate $\lim_{x \to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	$\to -1 \leftarrow$	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

$\lim_{x \to -1^-} f(x) \approx$ _____ $\lim_{x \to -1^+} f(x) \approx$ _____ $\lim_{x \to -1} f(x) \approx$ _____

Evaluate each limit. (2 points each)

9. $\lim_{x \to -4} \sqrt{x^2 + 9}$	**10.** $\lim_{x \to \frac{\pi}{6}} \left(-4\cos(2x)\right)$
11. $\lim_{x \to -1} f(x), \quad f(x) = \begin{cases} 2x + 3, & x < -1 \\ x^2 - 2, & x \geq -1 \end{cases}$	**12.** $\lim_{x \to 3} \dfrac{x + 2}{x - 3}$

Evaluate each limit. (3 points each)

13. $\lim\limits_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$	**14.** $\lim\limits_{x \to 6} \dfrac{\sqrt{x+3} - 3}{x - 6}$
15. $\lim\limits_{x \to -1} \dfrac{\frac{1}{x+3} - \frac{1}{2}}{x + 1}$	**16.** For $f(x) = \sqrt{x-2}$, find $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{2}$

Both sides of the line head to the open circle at $(1, 2)$.	Ignore the filled dot at $(1, -4)$.

2. $\boxed{-4}$

The filled dot at $x = 1$ is at height -4.	Function value.

3. $\boxed{\text{DNE}}$

$\lim\limits_{x \to -3^-} f(x) = 0, \lim\limits_{x \to -3^+} f(x) = -2$	Left: the parabola ends at the open circle $(-3, 0)$. Right: the line starts at the open circle $(-3, -2)$.
$0 \neq -2$, so $\lim_{x \to -3} f(x)$ does not exist.	The dot $(-3, 3)$ is $f(-3)$ and does not matter.

4. $\boxed{-\infty}$

Both sides of $x = 5$ fall without bound.	Same direction from both sides, so the two-sided answer is $-\infty$.

5. $\boxed{a = -7,\ a = -1}$

Height 0 is reached from both sides at $x = -7$ and $x = -1$.	Two-sided approach to $y = 0$.
Not $x = -3$ (left side only), not $x = 3$ (right side only; the left heads to 4), not $x = 7$ (graph ends there).	One-sided approaches and endpoints do not count.

6. $\boxed{-3}$

$\lim\limits_{x \to 1^-} [f(x)g(x)] = \left[\lim\limits_{x \to 1^-} f(x)\right]\left[\lim\limits_{x \to 1^-} g(x)\right]$	Limit of a product (works for one-sided limits).
$= (1)(-3) = -3$	f from the left heads to the open circle $(1, 1)$; g has a hole at $(1, -3)$.

7. $\boxed{\text{DNE}}$

$\lim\limits_{x \to 1^-} [f(x) + g(x)] = 1 + (-3) = -2$	Left: $f \to 1$, $g \to -3$.
$\lim\limits_{x \to 1^+} [f(x) + g(x)] = -2 + (-3) = -5$	Right: $f \to -2$ (filled dot), $g \to -3$.
$-2 \neq -5$, so $\lim_{x \to 1} [f(x) + g(x)]$ does not exist.	You cannot split the two-sided limit because $\lim_{x \to 1} f(x)$ does not exist; work each side.

8. $\boxed{\text{DNE (suggested)}}$

Left: 3.21, 3.0201, 3.002001	Heading to 3.
Right: $-0.8,\ -0.98,\ -0.998$	Heading to -1.
$\lim\limits_{x \to -1^-} f(x) \approx 3, \quad \lim\limits_{x \to -1^+} f(x) \approx -1$	The two trends approach different values, so the table suggests $\lim_{x \to -1} f(x)$ does not exist. A table is evidence, not proof.

9. $\boxed{5}$

$\lim\limits_{x \to -4} \sqrt{x^2 + 9} = \sqrt{\lim\limits_{x \to -4} (x^2 + 9)}$	Limit of a root (the inside is positive).
$= \sqrt{16 + 9} = \sqrt{25} = 5$	Substitute.

10. $\boxed{-2}$

$\lim_{x\to\frac{\pi}{6}}(-4\cos(2x)) = -4\cos\left(\frac{\pi}{3}\right)$	Substitute: $2\cdot\frac{\pi}{6}=\frac{\pi}{3}$.
$= -4\left(\frac{1}{2}\right) = -2$	Unit circle.

11. $\boxed{\text{DNE}}$

$\lim_{x\to-1^-} f(x) = \lim_{x\to-1^-}(2x+3) = -2+3 = 1$	Left of -1 uses $2x+3$.
$\lim_{x\to-1^+} f(x) = \lim_{x\to-1^+}\left(x^2-2\right) = 1-2 = -1$	Right uses x^2-2.
$1 \neq -1$, so $\lim_{x\to-1} f(x)$ does not exist.	Sides disagree.

12. $\boxed{\text{DNE}}$

Substitution produces $\frac{5}{0}$: a nonzero number over 0.	No finite limit. The limit does not exist; done.
Left: top near 5, bottom tiny negative, so $-\infty$. Right: $+\infty$.	Opposite directions, so $\lim_{x\to3}\frac{x+2}{x-3}$ does not exist.

13. $\boxed{3}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim_{x\to2}\frac{x^3-8}{x^2-4} = \lim_{x\to2}\frac{(x-2)\left(x^2+2x+4\right)}{(x-2)(x+2)}$	Difference of cubes; difference of squares.
$= \lim_{x\to2}\frac{x^2+2x+4}{x+2}$	Cancel $x-2$.
$= \frac{4+4+4}{2+2} = \frac{12}{4} = 3$	Substitute.

14. $\boxed{\frac{1}{6}}$

Substitution produces 0/0: indeterminate.	Multiply by the conjugate $\sqrt{x+3}+3$.
$\lim_{x\to6}\frac{\sqrt{x+3}-3}{x-6} = \lim_{x\to6}\frac{(x+3)-9}{(x-6)\left(\sqrt{x+3}+3\right)}$	Top is a^2-b^2. Do not FOIL; leave the bottom factored.
$= \lim_{x\to6}\frac{x-6}{(x-6)\left(\sqrt{x+3}+3\right)} = \lim_{x\to6}\frac{1}{\sqrt{x+3}+3}$	Cancel $x-6$.
$= \frac{1}{\sqrt{9}+3} = \frac{1}{6}$	Substitute.

15. $\boxed{-\frac{1}{4}}$

Substitution produces 0/0: indeterminate.	Multiply top and bottom by the LCD $2(x+3)$.
$\lim\limits_{x\to -1} \dfrac{\frac{1}{x+3}-\frac{1}{2}}{x+1} = \lim\limits_{x\to -1} \dfrac{2-(x+3)}{2(x+3)(x+1)}$	$2(x+3)\cdot\frac{1}{x+3}=2$, $2(x+3)\cdot\frac{1}{2}=x+3$.
$= \lim\limits_{x\to -1} \dfrac{-(x+1)}{2(x+3)(x+1)}$	$2-x-3=-x-1=-(x+1)$.
$= \lim\limits_{x\to -1} \dfrac{-1}{2(x+3)} = \dfrac{-1}{2(2)} = -\dfrac{1}{4}$	Cancel $x+1$; substitute.

16. $\boxed{\frac{1}{2\sqrt{x-2}}}$

$\lim\limits_{h\to 0} \dfrac{\sqrt{x+h-2}-\sqrt{x-2}}{h}$	$f(x+h)=\sqrt{(x+h)-2}$. Substituting $h=0$ produces 0/0.
$= \lim\limits_{h\to 0} \dfrac{(x+h-2)-(x-2)}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)}$	Multiply by the conjugate.
$= \lim\limits_{h\to 0} \dfrac{h}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)} = \lim\limits_{h\to 0} \dfrac{1}{\sqrt{x+h-2}+\sqrt{x-2}}$	Cancel h.
$= \dfrac{1}{2\sqrt{x-2}}$	Substitute 0 for h. Needs $x>2$.